

Tiki E-Commerce Data Analysis and Visualization

Final Report for Data Analysis and Visualization

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Abstract—This report investigates product-level behavior in the Vietnamese Tiki e-commerce fashion market. The analysis is structured around three central questions: whether Buy Now Pay Later (BNPL) functions as a financial lever for higher-value products, whether perfect 5.0-star ratings are less persuasive than slightly imperfect but crowd-validated ratings, and whether gender-oriented product groups differ consistently across shoes and bags. The study applies exploratory analysis, distributional assessment, non-parametric hypothesis testing, correlation analysis, and multiple linear regression. The findings suggest that BNPL is strongly associated with higher price levels and brand-level revenue structure, but it is not sufficient to guarantee higher unit sales. Rating performance depends heavily on review volume, which acts as a social-proof mechanism. Gender-oriented differences are visible in price structure, but sales velocity is more category-dependent and should be interpreted cautiously.

Index Terms—e-commerce analytics, exploratory data analysis, BNPL, rating paradox, social proof, gender-category analysis, Spearman correlation, multiple linear regression

I. INTRODUCTION

Digital marketplaces create a rich environment for observing how product attributes, customer trust, payment options, and category positioning interact with sales performance. In fashion-oriented e-commerce, these relationships are especially important because buyers often make decisions under uncertainty: they cannot physically inspect the product, they rely on ratings and reviews, and they may be influenced by delivery convenience or payment flexibility.

This study analyzes a Tiki product dataset from fashion-related categories. The objective is not merely to describe the dataset, but to develop a set of practical analytical arguments about consumer response in an online marketplace. Three themes guide the report. First, BNPL is examined as a possible financial lever. Second, the rating paradox is studied through the contrast between perfect ratings and highly rated products with stronger review volume. Third, gender-oriented behavior is examined across both shoes and bags to reduce the risk of drawing conclusions from a single category.

The report follows the structure of a full data analysis pipeline: data preparation, cleaning, exploratory analysis, probability distribution analysis, hypothesis testing, correlation analysis, and regression modeling. The figures referenced throughout the report are intended to support each theoretical claim and empirical interpretation.

II. DATA PREPARATION

A. Dataset Structure

The dataset consists of product listings from six fashion-related groups: men's bags, women's bags, men's shoes, women's shoes, backpacks and suitcases, and fashion accessories. Each product record includes variables such as price, original price, brand, rating, review count, delivery type, video availability, BNPL support, product age, and cumulative quantity sold.

To support category-level interpretation, each product is assigned a main category and a target-gender label. Bags and shoes are separated into male- and female-targeted groups, while backpacks, suitcases, and accessories are treated as unisex groups. This structure allows the analysis to distinguish product-category effects from gender-oriented effects.

TABLE I
INPUT PRODUCT GROUPS

Product group	Main category	Target group
Men bags	Bags	Male
Women bags	Bags	Female
Men shoes	Shoes	Male
Women shoes	Shoes	Female
Backpacks and suitcases	Backpacks - Suitcases	Unisex
Fashion accessories	Accessories	Unisex

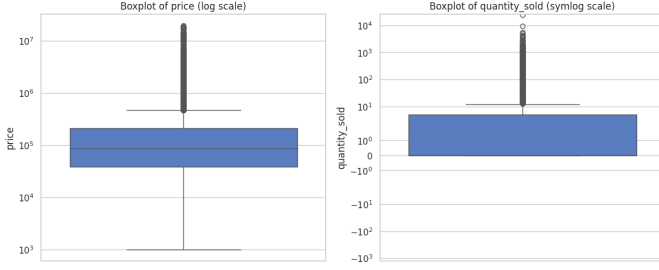
B. Cleaning Principles

Data cleaning is guided by analytical validity. Missing brand values are treated as OEM products because marketplace datasets often include unbranded or seller-branded goods. Missing delivery information is assigned to the standard non-platform-fulfilled case. Missing sales quantity is interpreted as zero cumulative sales, which is consistent with product records that have not yet generated orders.

Duplicate product records are removed using the product identifier. Products with zero price are excluded from price and revenue analysis because they are more likely to represent gifts, promotional items, or non-standard listings. This removal prevents price-based metrics from being distorted by records that do not represent normal selling activity.

Outliers are detected with the interquartile range method. However, extreme values are not removed automatically. In an e-commerce setting, very expensive products and best-selling products may be genuine and meaningful. For this reason,

logarithmic or symmetric-logarithmic visualization scales are used to preserve information while making distributions readable.



C. Derived Analytical Measures

Several derived measures are used to support interpretation. Sales velocity measures average cumulative sales per day and helps reduce bias caused by product age. Discount percentage captures the relative price reduction from original price to current price. Revenue proxy, defined as price multiplied by cumulative quantity sold, approximates the economic contribution of each product. Price segment, review tier, and rating tier are used to compare products within more interpretable groups.

These measures convert raw listing attributes into behavioral indicators. They do not claim to measure causality, but they allow systematic comparison across BNPL status, rating groups, gender-oriented categories, and product segments.

III. EXPLORATORY ANALYSIS

A. Category Composition

Before evaluating the three research themes, the category structure must be understood. Product categories differ in SKU count, price range, review behavior, and sales dynamics. A direct comparison without considering category composition can produce misleading conclusions. For example, accessories may have many listings and low prices, while shoes and bags may show more pronounced gender-oriented segmentation.

The category and gender-composition figures provide this baseline. They show how the dataset is distributed across product groups and whether male, female, and unisex products appear in comparable proportions.

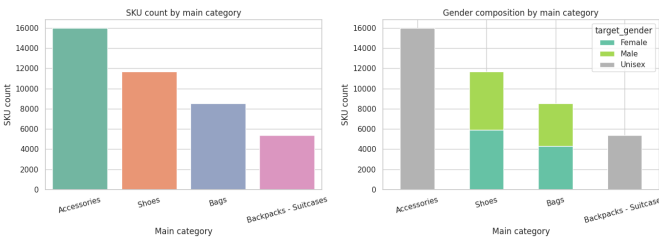


Fig. 2. Product count by category and gender composition by category.

B. BNPL as a Financial Lever

BNPL is theoretically expected to reduce the immediate payment burden for buyers. In online shopping, this can be especially relevant for products that are desirable but relatively expensive. The first hypothesis is therefore that BNPL should be more closely associated with higher-value products than with low-cost impulse purchases.

To evaluate this idea, products are grouped by cumulative sales tier and the BNPL coverage rate is visualized for each tier. The stacked bar chart is used to examine whether BNPL appears only among products that already sell well, or whether it is distributed more broadly across sales-performance groups.

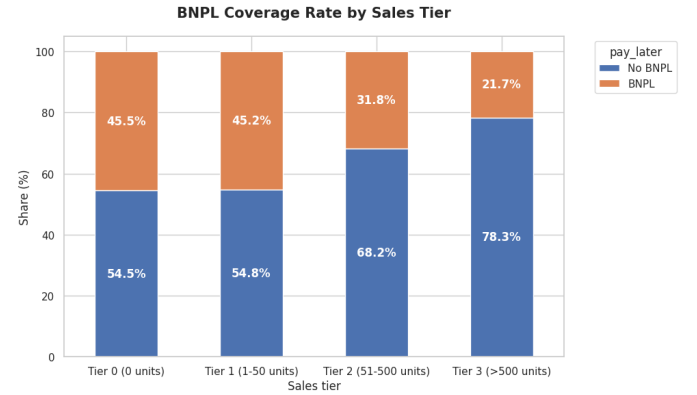


Fig. 3. BNPL coverage rate across cumulative sales tiers.

Revenue proxy is then used to compare the economic contribution of BNPL and non-BNPL products. A pie chart summarizes the share of cumulative revenue proxy, while a log-scale boxplot compares the revenue proxy generated by individual products. This is important because unit sales alone can understate the contribution of higher-priced products.

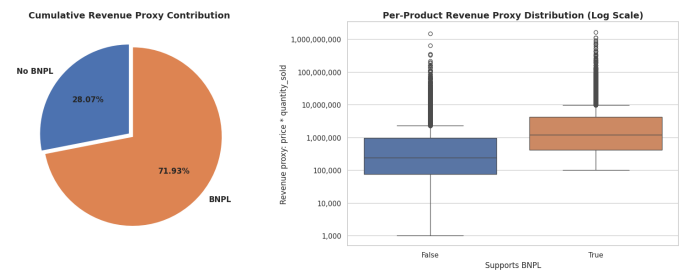


Fig. 4. Revenue-proxy contribution and per-product revenue distribution by BNPL status.

The price-distribution boxplot provides a direct view of the relationship between BNPL and product value. If the BNPL group has a higher median price and a wider upper range, then BNPL is better understood as a financing mechanism attached to higher-value products.

Trust and brand structure provide a second layer of explanation. A rating-revenue scatter plot examines whether high rating alone corresponds to stronger revenue proxy, while

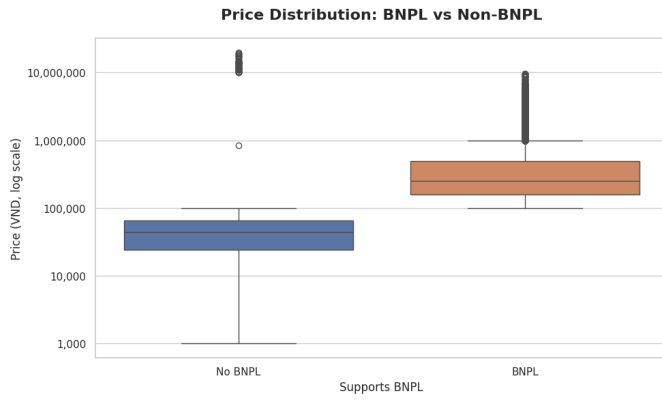


Fig. 5. Price distribution of BNPL and non-BNPL products.

brand-level visualizations show whether BNPL support is concentrated among commercially stronger brands.

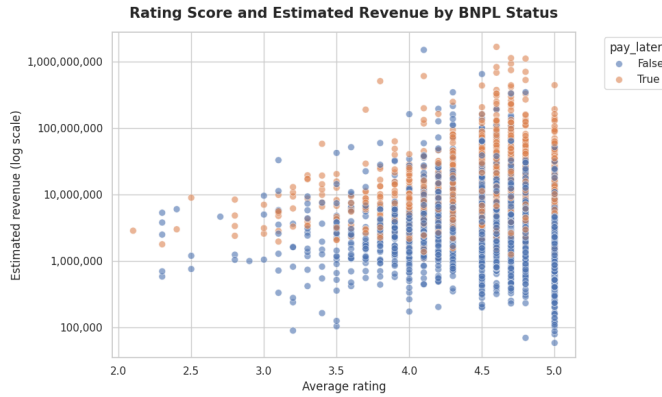


Fig. 6. Average rating and estimated revenue by BNPL status.

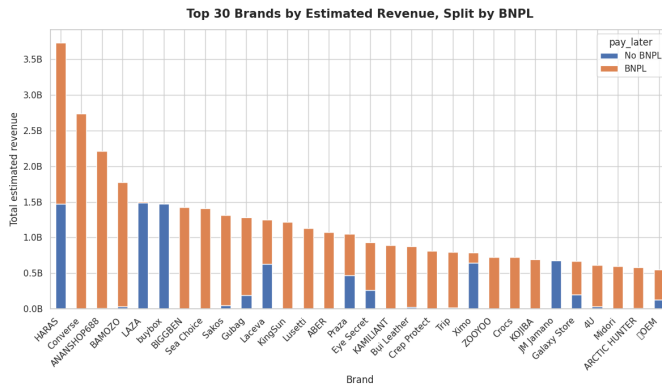


Fig. 7. Top 30 brand-level estimated revenue split by BNPL status.

Taken together, these figures support a cautious interpretation. BNPL is clearly related to product value and commercial positioning, but higher BNPL availability should not be treated as direct proof of higher unit sales. Payment flexibility may improve purchase feasibility, yet demand still depends on brand, category, review trust, and price sensitivity.

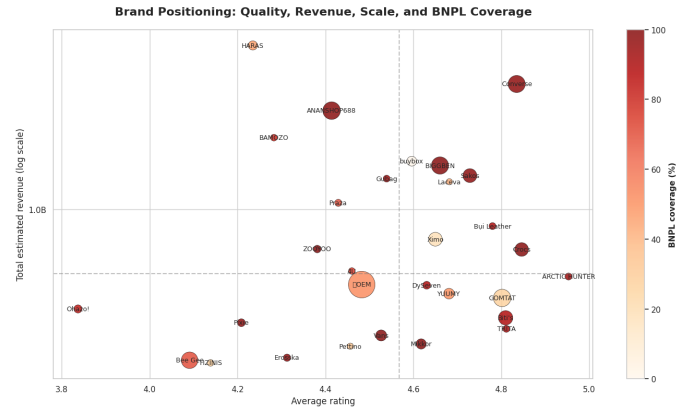


Fig. 8. Brand positioning by average rating, estimated revenue, SKU scale, and BNPL coverage.

C. The Rating Paradox

Online buyers often use ratings as a shortcut for trust. However, a perfect 5.0-star score can be ambiguous when it is based on very few reviews. A slightly lower rating, such as 4.7–4.9 stars, may be more persuasive if it is supported by many customers. This creates the rating paradox: the apparently perfect product is not necessarily the most credible product.

The analysis first compares products with perfect ratings against products with 4.7–4.9-star ratings. The purpose is to separate the symbolic appeal of a perfect rating from the stability created by review volume. A product with one 5.0-star review is mathematically perfect but socially fragile, while a product with hundreds of 4.8-star reviews is less perfect but more validated.

TABLE II
CONCEPTUAL INTERPRETATION OF RATING GROUPS

Rating group	Trust signal	Main limitation
5.0 stars	Perfect score	Often low review volume
4.7–4.9 stars	High score with social proof	Slightly imperfect score

The review-tier barplot is central to this argument. It compares median sales between rating groups within the same review-volume range. This design prevents review volume from overwhelming the interpretation of rating score. If the 4.7–4.9 group performs strongly within comparable review tiers, the result supports the idea that buyers value credible social proof more than a perfect but sparse score.

The density plot of log-transformed sales provides a distributional view of the same phenomenon. Because sales are skewed, logarithmic transformation makes the comparison more interpretable. The chart shows whether the two rating groups occupy different sales regions among products with comparable review volume.

The Bayesian trust illustration strengthens the theoretical interpretation. A sparse perfect rating is unstable because one negative review can sharply reduce its score. A high rating

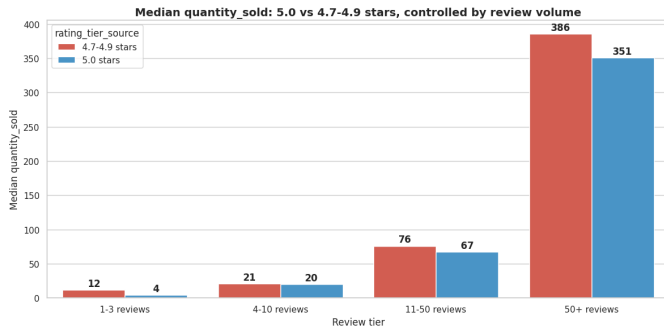


Fig. 9. Median sales of 5.0-star and 4.7–4.9-star products by review tier.

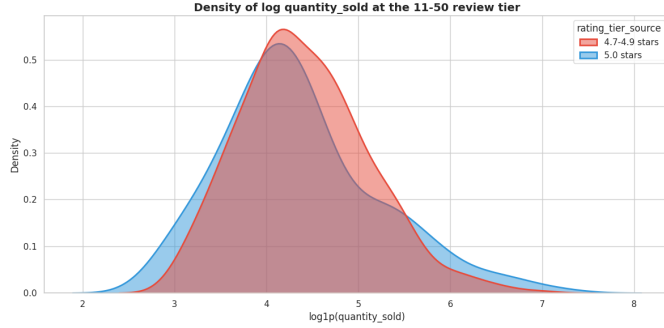


Fig. 10. Log-sales distribution for perfect and crowd-validated rating groups.

supported by many reviews is more robust because each new review has a smaller impact on the overall rating. This explains why review volume should be interpreted as a credibility mechanism rather than a secondary metric.

=== Bayesian stability illustration ===
 Global mean rating: 4.41; confidence anchor: 9.94 reviews
 Sparse perfect-rating product:
 Raw rating: 5.0; reviews: 1; Bayesian score: 4.47
 Crowd-validated high-rating product:
 Raw rating: 4.7; reviews: 105; Bayesian score: 4.68
 One 1-star review changes a 5.0-star / 1-review product to 3.0.
 One 1-star review changes a 4.8-star / 100-review product to 4.76.

Fig. 11. Bayesian trust score and robustness of rating groups.

D. Gender-Category Behavior

Gender-oriented product analysis must be handled carefully. If only shoes are examined, observed differences may reflect the shoe category rather than general gender-oriented behavior. For this reason, both shoes and bags are analyzed. This design allows the study to check whether patterns repeat across categories or remain category-specific.

For shoes, price distribution and sales velocity are examined together. A price boxplot shows whether male- and female-targeted shoes occupy different price ranges. A price–sales velocity scatter plot then shows whether demand concentrates in particular price regions.

Separate Spearman heatmaps for male- and female-targeted shoes summarize behavioral relationships among price, re-

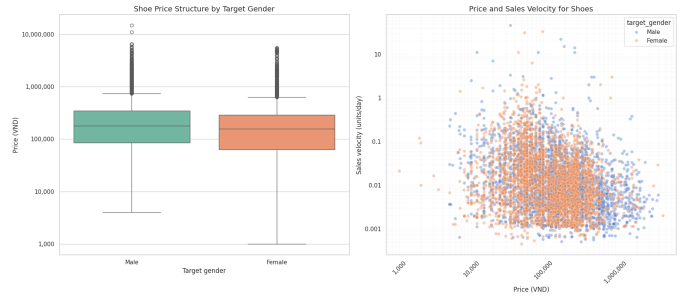


Fig. 12. Shoe price structure and sales velocity by target group.

view count, rating, discount, OEM status, product age, video availability, and sales velocity. Spearman correlation is used because marketplace variables are non-normal and contain many outliers.

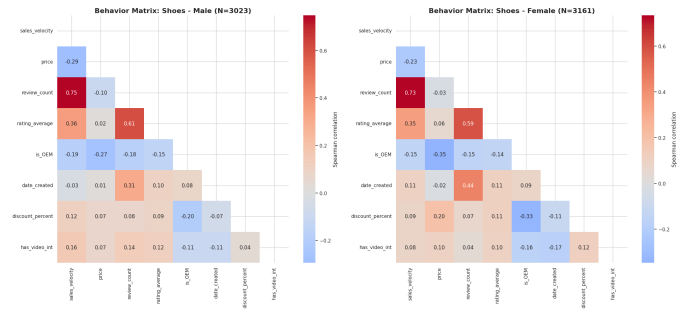


Fig. 13. Behavioral Spearman matrices for male- and female-targeted shoes.

The review-threshold chart evaluates whether accumulating reviews is associated with stronger sales velocity. The weighted price-density plot shows where active shoe demand concentrates. Together, these figures connect the gender-category discussion with the broader theory of social proof and price sensitivity.

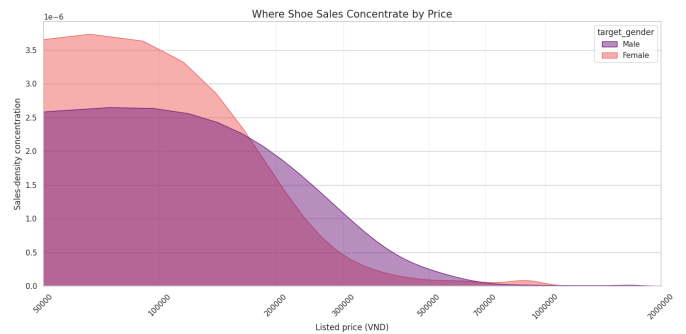


Fig. 14. Review threshold and price concentration in shoe sales.

The bag analysis mirrors the shoe analysis. SKU count, price distribution, and sales performance are examined first, followed by separate Spearman heatmaps for male- and female-targeted bags. This paired structure allows the report to distinguish consistent gender-category tendencies from category-specific patterns.

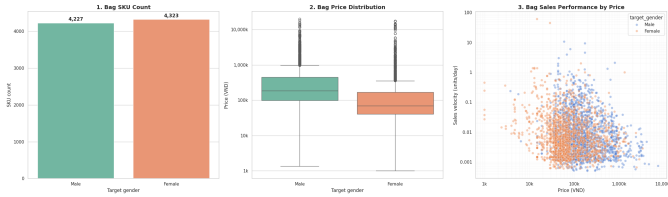


Fig. 15. Bag SKU count, price structure, and sales performance by target group.

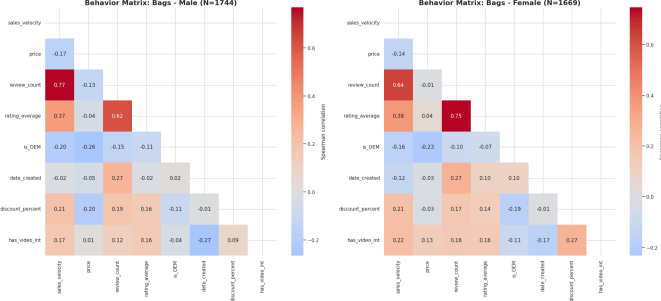


Fig. 16. Behavioral Spearman matrices for male- and female-targeted bags.

Product video is treated as an information-rich signal. In theory, video can reduce uncertainty by showing product form, texture, or usage. The video-effect chart compares sales velocity for products with and without video across shoes and bags, while non-parametric tests evaluate whether the observed differences are statistically meaningful.

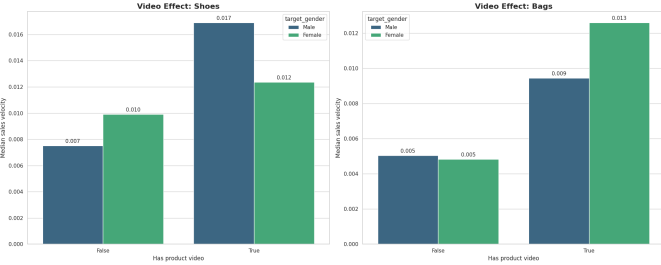


Fig. 17. Median sales velocity by product-video availability across shoes and bags.

IV. PROBABILITY DISTRIBUTION ANALYSIS

Price is selected for distributional analysis because it directly affects all three themes. BNPL is expected to interact with price level, rating trust may matter differently across price ranges, and gender-category comparisons are partly expressed through price structure.

The raw price distribution is strongly right-skewed, which is typical of marketplace data. Most products are concentrated in lower and middle price ranges, while a small number of expensive products form a long upper tail. A normal distribution is therefore inappropriate for raw price. Log transformation provides a more meaningful representation because prices often vary multiplicatively.

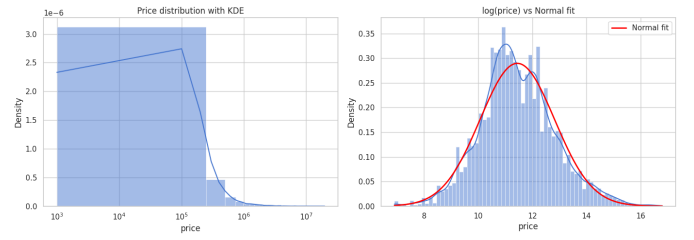


Fig. 18. Raw and log-transformed price distributions.

The Q-Q plot and standardized log-price histogram assess how closely log-price approximates a normal distribution. Even if formal tests reject exact normality, the visual analysis remains useful because it justifies log-scale visualization and log-based modeling decisions.

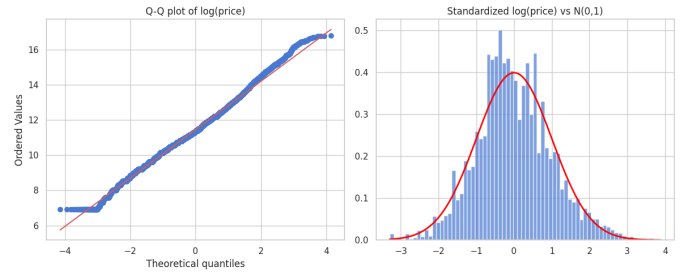


Fig. 19. Normality assessment for log-transformed price.

V. HYPOTHESIS TESTING

A. BNPL and Sales Performance

The BNPL hypothesis asks whether products with payment flexibility achieve higher sales performance. Because quantity sold is skewed and includes many small values, a Mann-Whitney U test is preferred over a parametric mean-comparison test. The theoretical expectation is mixed: BNPL may reduce payment friction, but it is also more common among higher-priced products, which may naturally sell fewer units.

The result should therefore be interpreted through both statistical significance and business meaning. A difference between BNPL and non-BNPL products does not automatically imply that BNPL causes higher unit sales. It may instead indicate that BNPL is attached to a different product segment.

B. Rating and Social Proof

The rating hypothesis evaluates whether the 4.7–4.9-star group can outperform the 5.0-star group after review volume is considered. The theoretical basis is that buyers may distrust overly perfect ratings when they are supported by too few reviews. Review volume functions as social proof and stabilizes rating credibility.

The test focuses on products with sufficient review count so that the comparison is not dominated by sparse-rating products. A significant difference would support the argument that credibility depends on both rating score and the amount of customer confirmation behind that score.

C. Gender-Category Differences

The gender-category hypothesis asks whether differences between male- and female-targeted products are consistent across shoes and bags. The test compares price, discount, review count, sales velocity, and delivery support within each category. This structure avoids treating category-specific patterns as universal gender behavior.

The interpretation emphasizes caution. Stronger price differences do not necessarily imply stronger sales differences. Sales velocity is affected by price range, reviews, product media, delivery conditions, and category-level demand.

VI. CORRELATION ANALYSIS

Correlation analysis provides a compact view of relationships among marketplace variables. Spearman correlation is used because the variables are skewed, non-normal, and sensitive to outliers. The bag heatmaps compare male- and female-targeted products separately, allowing the relationship structure to be interpreted within each gender-category group.

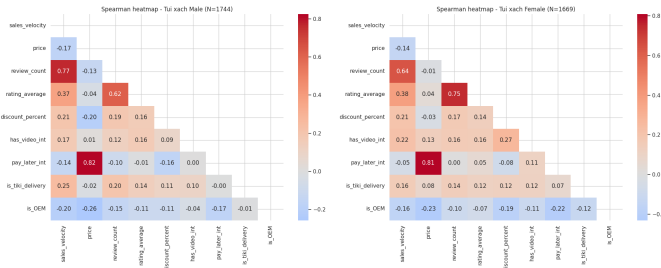


Fig. 20. Spearman correlation heatmaps for gender-targeted bags.

Scatter plots are used to make selected relationships visible. Review count versus quantity sold illustrates the practical importance of social proof. Price versus sales velocity shows how demand changes across price ranges and BNPL status. Log and symlog scales are necessary because most products cluster at low values while a small number of products dominate the upper tail.

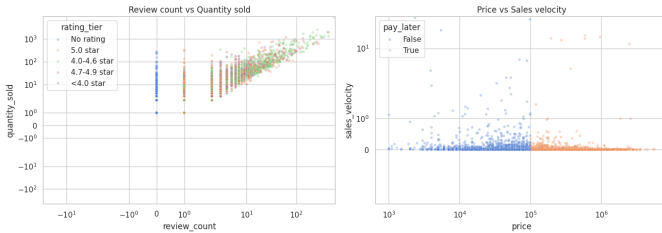


Fig. 21. Key scatter plots for review count, sales quantity, price, and sales velocity.

The correlation results reinforce the earlier themes. Review volume is closely related to sales performance, while BNPL is strongly connected to product price. These relationships support the view that trust signals and financial accessibility operate together rather than independently.

VII. MULTIPLE LINEAR REGRESSION

Multiple linear regression is used to evaluate how several product attributes jointly relate to sales performance. The dependent variable is log-transformed cumulative quantity sold, which reduces skewness and makes the model less dominated by extreme best-sellers.

The explanatory variables represent price, review volume, rating, discount, product video, BNPL support, delivery support, OEM status, product age, category, gender orientation, and price segment. This design allows the model to compare the relative strength of trust, price, product information, payment support, and category positioning.

Residual diagnostics evaluate whether the model leaves systematic patterns unexplained. The residual-vs-fitted plot checks for non-linearity, the Q-Q plot assesses residual normality, and the residual histogram shows whether skewness remains after transformation.

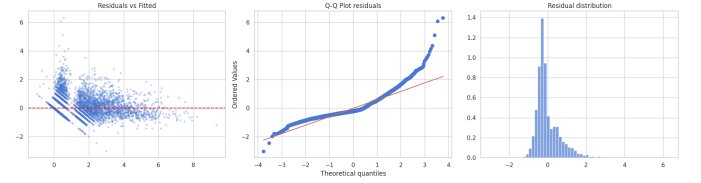


Fig. 22. Residual diagnostics for the multiple linear regression model.

The regression interpretation is consistent with the exploratory findings. Review volume is expected to be one of the strongest positive predictors of sales performance, while price often has a negative association with unit sales. BNPL may be positive in some specifications, but its effect must be interpreted carefully because BNPL is strongly tied to price segment and product positioning.

VIII. DISCUSSION

The results show that marketplace behavior cannot be explained by a single visible attribute. BNPL, rating, review volume, price, category, and delivery signals interact with one another. A product may support BNPL because it is expensive, not necessarily because BNPL directly increases its unit sales. A product may have a perfect rating because it has too few reviews, not because it has stronger market trust. A gender-oriented price difference may reflect category structure rather than a universal difference in buyer behavior.

The strongest conceptual pattern across the analysis is the role of trust. Review volume appears repeatedly in the rating paradox, correlation analysis, and regression modeling. This suggests that in online shopping, social proof is not merely a descriptive metric; it is a central mechanism through which buyers reduce uncertainty.

IX. DISCUSSION

The results show that marketplace behavior cannot be explained by a single visible attribute. BNPL, rating, review volume, price, category, and delivery signals interact with one another. A product may support BNPL because it is expensive,

TABLE III
SUMMARY OF NON-PARAMETRIC HYPOTHESIS TESTS

Hyp.	Comparison	Key statistics	p-value	Conclusion
H1	BNPL vs non-BNPL products by quantity_sold	Mann-Whitney U = 201,310,778.5; mean BNPL = 11.01; mean non-BNPL = 23.19; median = 0 for both groups; mean uplift ratio = 0.47x	6.421×10^{-27}	Significant difference; BNPL does not imply higher unit sales.
H2	5.0-star products vs 4.7–4.9-star products after review control	Mann-Whitney U = 94,767.0; median 5.0-star = 73; median 4.7–4.9-star = 102	1.991×10^{-5}	Significant difference; the crowd-validated group performs better.
H3a	Male vs female products by price, within category	Shoes: male median = 160,000, female median = 129,000, difference = 31,000; Bags: male median = 141,900, female median = 65,000, difference = 76,900	Shoes: 1.262×10^{-18} ; Bags: 2.778×10^{-108}	Significant price differences in both categories.
H3b	Male vs female products by sales velocity, within category	Shoes: male median = 0.008210, female median = 0.010165, difference = -0.001955; Bags: male median = 0.006064, female median = 0.005576, difference = 0.000487	Shoes: 5.347×10^{-4} ; Bags: 3.478×10^{-1}	Sales-velocity difference is category-dependent.

TABLE IV
SELECTED COEFFICIENTS IN THE MULTIPLE LINEAR REGRESSION MODEL

Predictor	Coef.	t-stat.	p-value	Interpretation
log_review_count	1.1654	184.371	< 0.001	Strongest positive predictor; review volume is the main social-proof signal.
log_price	-0.1472	-26.067	< 0.001	Higher price is associated with lower unit sales after controlling for other variables.
rating_average	0.0825	30.561	< 0.001	Higher ratings are positively associated with sales, but the effect is smaller than review volume.
discount_percent	0.0016	6.066	< 0.001	Discount depth has a positive but small association with sales.
has_video_int	0.0999	8.329	< 0.001	Product video is associated with higher sales performance.
pay_later_int	0.0337	0.648	0.517	BNPL is not statistically significant after controls.
is_tiki_delivery	0.3442	12.379	< 0.001	Platform delivery is positively associated with sales performance.
is_OEM	-0.1458	-16.196	< 0.001	OEM or unbranded products are associated with lower sales performance.
main_category_Shoes	0.3466	19.002	< 0.001	Shoes have higher expected log sales than the reference category.
target_gender_Male	-0.0131	-1.275	0.202	Male-targeted status is not statistically significant after controls.
price_segment_>1m	0.2077	3.625	< 0.001	Products above 1 million VND retain a positive association after segment controls.

not necessarily because BNPL directly increases its unit sales. A product may have a perfect rating because it has too few reviews, not because it has stronger market trust. A gender-oriented price difference may reflect category structure rather than a universal difference in buyer behavior.

The strongest conceptual pattern across the analysis is the role of trust. Review volume appears repeatedly in the rating paradox, correlation analysis, and regression modeling. This suggests that in online shopping, social proof is not merely a descriptive metric; it is a central mechanism through which buyers reduce uncertainty.

X. CONTRIBUTIONS

The project was completed collaboratively by three members. Table V summarizes the main individual responsibilities and estimated completion levels. The percentages reflect task completion within the assigned scope rather than grading weights.

XI. SELF-EVALUATION

The project requirements were completed at a high level. The final work includes a complete data pipeline, exploratory analysis by research theme, probability distribution analysis, hypothesis testing, Spearman correlation analysis, and multiple

TABLE V
INDIVIDUAL CONTRIBUTIONS AND COMPLETION LEVELS

Student ID	Name	Main responsibilities	Completion
52300135	Le Quang Hong Ngoc	Data understanding, data cleaning review, final category, presentation preparation and report writing support.	100%
52300138	Nguyen Tri Nhan	Gender-oriented analysis, interpretation of shoes and bags behavior, notebook integration, pipeline synchronization, visualization organization, LaTeX/IEEE report preparation.	100%
52300143	Nguyen Lai Vu Phong	BNPL research design, Pay Later analysis, revenue-proxy interpretation, brand-level BNPL visualization, hypothesis framing.	100%

linear regression. The three research questions are addressed with both visual and statistical evidence.

TABLE VI
SELF-EVALUATION OF PROJECT COMPLETION

Requirement	Completion
Data loading and schema checking	100%
Data cleaning and feature engineering	100%
EDA by three research themes	100%
Probability distribution analysis	100%
Hypothesis testing	100%
Correlation analysis	100%
Multiple linear regression	100%
IEEE report and presentation support	100%

REFERENCES

- [1] Tiki.vn, “Vietnamese e-commerce product listings,” dataset used for educational data analysis.

XII. CONCLUSION

This report satisfies the project requirements by presenting a complete analysis of Tiki fashion-related product data. The approach begins with dataset preparation and cleaning, continues with feature engineering and exploratory analysis, and then applies distributional analysis, non-parametric hypothesis testing, Spearman correlation, and multiple linear regression. The workflow provides both descriptive and inferential evidence for the three research themes.

The experimental results show that BNPL is associated with higher-value products and brand-level revenue structure, but it should not be interpreted as automatic evidence of higher unit sales. The rating analysis supports the rating paradox: a 5.0-star score is not always the strongest trust signal when review volume is low, while slightly lower but more crowd-validated ratings may be more credible to buyers. The gender-category analysis shows that including both shoes and bags leads to a more careful interpretation: price differences are visible, but sales velocity depends on multiple category and product signals.

Overall, the project is considered fully completed. The final notebook, report, and presentation materials cover the required analytical components and provide a coherent interpretation of the dataset. Future research can improve the analysis by adding time-series sales data, causal methods for BNPL and video effects, more detailed product text features, and category-specific predictive models.